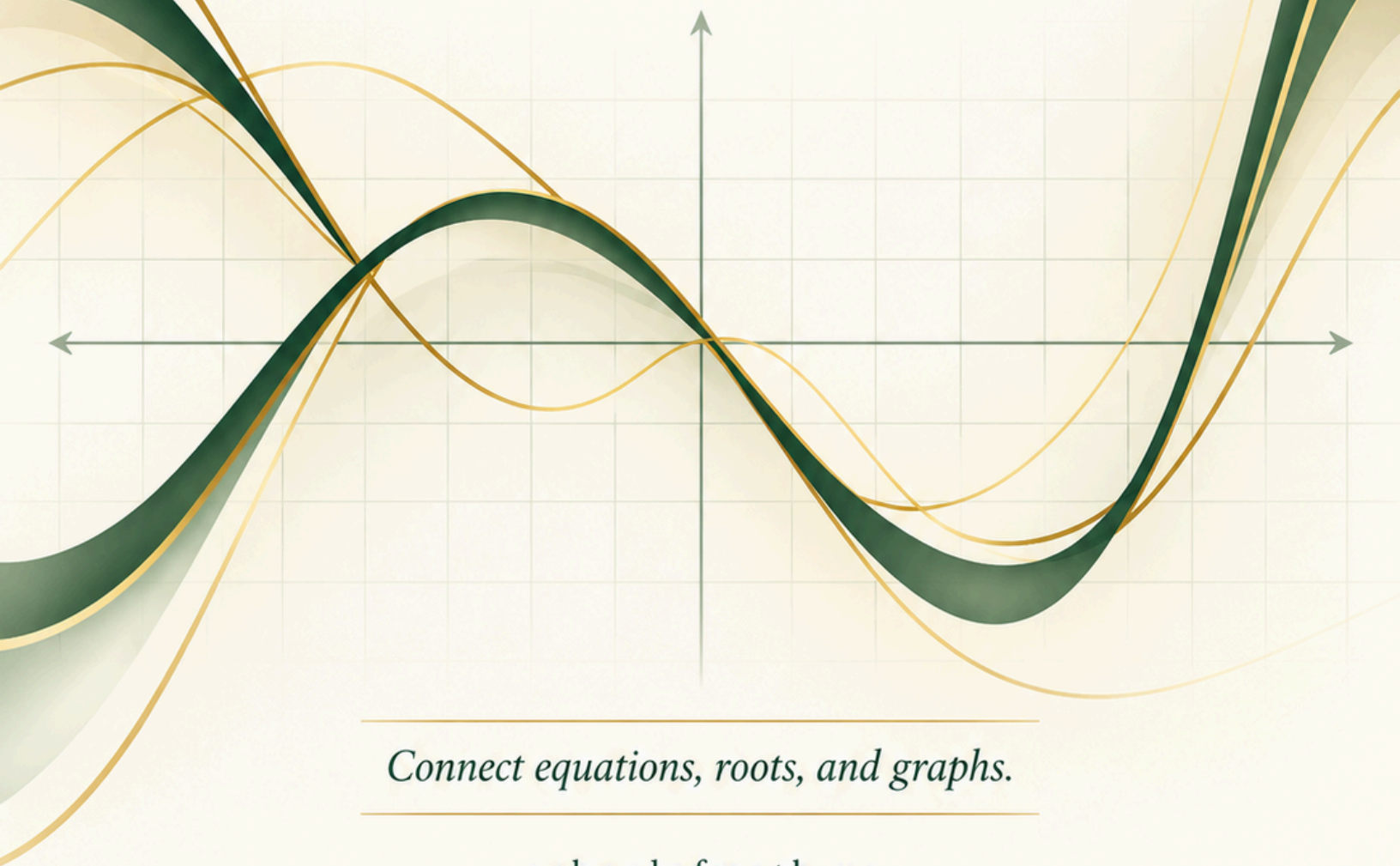


SCHOOL
OF MATH

ALGEBRA II • CLASS 12

Zeros of Polynomials

The Rational Root Theorem
& End Behavior



Connect equations, roots, and graphs.

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Explore
Reason
Solve
Grow

What Is a Zero of a Polynomial?

Same
Ideas.
Brighter
Futures.

A SINGLE VALUE. A BIG CONNECTION.



DEFINITION

A **zero** of a polynomial is a value of x that makes $P(x) = 0$.



VOCABULARY

Zeros are also called **roots** or **solutions**.



GRAPH CONNECTION

On the graph of $y = P(x)$, the zeros are the **x -intercepts**.



FACTOR CONNECTION

If $x = a$ is a zero, then $(x - a)$ is a factor of the polynomial.



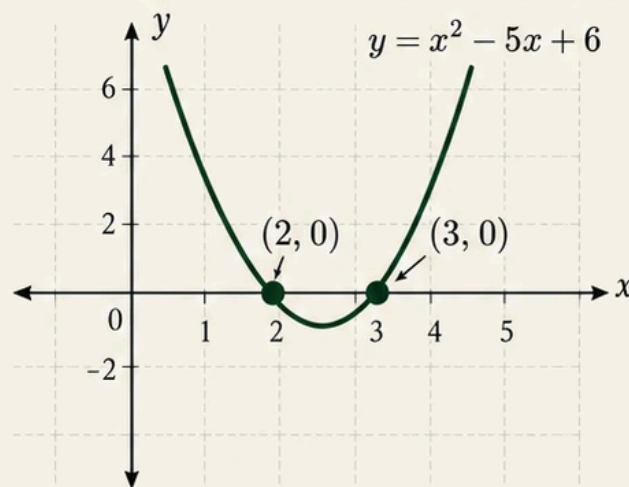
EXAMPLE

$$\begin{aligned} P(x) &= x^2 - 5x + 6 \\ &= (x - 2)(x - 3) \end{aligned}$$

Zeros: $x = 2$ and $x = 3$



GRAPH OF THE POLYNOMIAL



Zeros connect equations, factors, and graphs.



Image 1 practice: What is a zero of a polynomial?

Q1. For $P(x) = x^2 + x - 2$, decide whether $x = -2$ is a zero.

Q2. A polynomial has zeros at -4 and 5 . Write the corresponding linear factors and the two intercepts.

Q3. For $R(x) = (x - 6)(x + 3)$, find the zeros and the y -intercept. Explain why the intercept is not another zero.

Q4. The polynomial $P(x) = x^2 + ax - 12$ has a zero at 3 . Find a and the other zero.

Q5. A quadratic has zeros -2 and 7 , and its graph passes through $(0, 28)$. Find the quadratic and verify both zeros.

Finding Zeros by Factoring

Solve polynomial equations by factoring and using the Zero Product Property.

Key Idea

If a product is zero, then at least one of the factors is zero.
If $ab = 0$, then $a = 0$ or $b = 0$.

Example 1

Find the zeros of $x^2 - 5x + 6 = 0$.

- 1 Factor the polynomial.
$$x^2 - 5x + 6 = (x - 2)(x - 3)$$
- 2 Use the Zero Product Property.
$$x - 2 = 0 \quad \text{or} \quad x - 3 = 0$$
- 3 Solve for x .
$$x = 2 \quad \text{or} \quad x = 3$$

Zeros: 2 and 3

Example 2

Find the zeros of $2x^2 + 7x - 4 = 0$.

- 1 Factor the polynomial.
$$2x^2 + 7x - 4 = (2x - 1)(x + 4)$$
- 2 Use the Zero Product Property.
$$2x - 1 = 0 \quad \text{or} \quad x + 4 = 0$$
- 3 Solve for x .
$$2x = 1 \Rightarrow x = \frac{1}{2} \quad x = -4$$

Zeros: $\frac{1}{2}$ and -4



A Bit of History

René Descartes (1596–1650) helped establish the connection between algebra and graphs, showing that the zeros of a polynomial correspond to the x-intercepts of its graph.

Key Takeaway

Factoring turns a polynomial equation into simpler equations. By setting each factor equal to zero, we can find all the zeros.

Image 2 practice: Find zeros by factoring

Q6. Find all zeros of $x^2 - 7x + 10$.

Q7. Find all zeros of $3x^2 - 5x - 2$.

Q8. Find every zero of $x^3 + 4x^2 - 16x - 64$ by grouping.

Q9. Find all zeros of $2x^4 - 10x^2 + 8$, including multiplicities.

Q10. Find all real and nonreal zeros of $x^4 + x^2 - 20$. Which are graph intercepts?

The Rational Root Theorem

Learn how to generate all possible rational zeros using factors of the constant term and the leading coefficient.

Theorem

If $p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ has rational zeros in lowest terms $\frac{p}{q}$, then p is a factor of a_0 and q is a factor of a_n .

How to Use It

- 1 List the factors of the constant term a_0 .
- 2 List the factors of the leading coefficient a_n .
- 3 Form all possible rational zeros $\frac{p}{q}$ (in lowest terms).
- 4 Test the possible zeros using substitution or synthetic division.

Key Point



The theorem gives a list of possible rational zeros. You still need to test which ones are actually zeros.

Example 1

Find all possible rational zeros of

$$f(x) = 2x^3 - 5x^2 - x + 6.$$

- 1 Factors of the constant term (6):
 $\pm 1, \pm 2, \pm 3, \pm 6$
- 2 Factors of the leading coefficient (2):
 $\pm 1, \pm 2$
- 3 Possible rational zeros:
 $\pm 1, \pm 2, \pm 3, \pm 6, \pm \frac{1}{2}, \pm \frac{3}{2}$

Example 2

Find all possible rational zeros of

$$g(x) = 3x^4 + 6x^3 - 9x + 12.$$

- 1 Factors of the constant term (12):
 $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12$
- 2 Factors of the leading coefficient (3):
 $\pm 1, \pm 3$
- 3 Possible rational zeros:
 $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12$
 $\pm \frac{1}{3}, \pm \frac{2}{3}, \pm \frac{4}{3}$



A Bit of History

The Rational Root Theorem has been used for centuries, and Carl Friedrich Gauss (1777–1855) made major contributions to the study of polynomial equations and their roots.

Why It Matters

The Rational Root Theorem helps us find the candidate rational zeros of a polynomial, making it much easier to solve equations and analyze graphs.



Image 3 practice: Use the Rational Root Theorem

Q11. List all possible rational zeros of $x^3 + 2x^2 - 5x - 10$. You do not need to solve the polynomial.

Q12. List all distinct rational-zero candidates for $4x^3 - 3x^2 - 6x + 2$.

Q13. List all distinct rational-zero candidates for $6x^3 - 5x^2 + 2x - 4$.

Q14. Explain how to apply the Rational Root Theorem to $3x^4 - 7x^3 + 2x^2$, then find all its zeros.

Q15. List and test the rational candidates for $x^3 - 3x^2 - 2x + 6$. Then find the irrational zeros that the theorem does not list.

Testing Possible Rational Zeros

Use substitution or synthetic division to determine which possible rational roots are actual zeros.

Key Idea

The Rational Root Theorem gives a list of possible rational zeros. To find which ones are actual zeros, test each by substituting into the polynomial or by using synthetic division. If $f(r) = 0$, then r is a zero.

Example 1

Determine which of the possible rational zeros are actual zeros of $f(x) = x^3 - 4x^2 - x + 4$.

Possible rational zeros: $\pm 1, \pm 2, \pm 4$

- 1 Test using substitution (try $x = 1$).

$$\begin{aligned} f(1) &= 1^3 - 4(1)^2 - 1 + 4 \\ &= 1 - 4 - 1 + 4 \\ &= 0 \end{aligned}$$

Since $f(1) = 0$, $x = 1$ is a zero.

- 2 Check another value ($x = -2$).

$$\begin{aligned} f(-2) &= (-2)^3 - 4(-2)^2 - (-2) + 4 \\ &= -8 - 16 + 2 + 4 \\ &= -18 \neq 0 \end{aligned}$$

So $x = -2$ is not a zero.

Conclusion: $x = 1$ is a zero
(we would test the remaining values similarly).

Example 2

Use synthetic division to test whether $x = 2$ is a zero of $f(x) = 2x^3 - 3x^2 - 8x + 4$.

- 1 Set up synthetic division with 2.

$$\begin{array}{r|rrrr} 2 & 2 & -3 & -8 & 4 \\ & & 4 & 2 & -12 \\ \hline & 2 & 1 & -6 & -8 \end{array}$$

The remainder is -8 , which is not zero.
So $x = 2$ is not a zero.

- 2 Now test $x = -1$.

$$\begin{array}{r|rrrr} -1 & 2 & -3 & -8 & 4 \\ & & -2 & 5 & 3 \\ \hline & 2 & -5 & -3 & 0 \end{array}$$

The remainder is 0, so $x = -1$ is a zero.

Conclusion: $x = -1$ is a zero of $f(x)$.



Why It Matters

Testing possible rational zeros helps us find the actual roots of a polynomial, which can then be used to factor the polynomial and analyze its graph.



Tip

If a value does not work, move to the next one on the list. Once you find a zero, you can use synthetic division to reduce the degree and find the remaining zeros.

Poster correction: In the second example, the displayed polynomial is $f(x) = 2x^3 - 3x^2 - 8x + 4$. Its value at -1 is $-2 - 3 + 8 + 4 = 7$, not 0. The final synthetic-division entry should be 7, so -1 is **not** a zero. The test at 2, which gives -8 , is correctly rejected.

Image 4 practice: Test possible zeros

Q16. For $f(x) = x^3 - 2x^2 - 5x + 6$, test $x = 1$ and $x = 2$. Which corresponding linear factor is valid?

Q17. Test whether $x = \frac{1}{2}$ is a zero of $2x^3 + x^2 - 13x + 6$.

Q18. Find b so that dividing $P(x) = 2x^3 + bx^2 - 7x + 3$ by $x - 2$ leaves remainder 5. Is 2 a zero?

Q19. For $P(x) = x^3 - 6x^2 + 11x - 6$, test 1, 2, and 3. Explain why these tests give a complete factorization, not just a guess.

Q20. The polynomial $P(x) = x^3 + ax^2 + bx + 12$ has zeros at 2 and -3 . Find a , b , and its third zero.

Using Synthetic Division to Reduce a Polynomial

Divide by a known factor $x - r$ and reduce the degree of the polynomial to find the remaining zeros.

Key Idea

If r is a zero of a polynomial $f(x)$, then $x - r$ is a factor of $f(x)$. We can use synthetic division to divide $f(x)$ by $x - r$, obtain a quotient polynomial of one degree less, and then find the remaining zeros.

Example 1

Given that 2 is a zero of $f(x) = x^3 - 4x^2 - x + 6$, use synthetic division to find the quotient polynomial and the remaining zeros.

- 1 Set up synthetic division with $r = 2$.

$$\begin{array}{r|rrrr} 2 & 1 & -4 & -1 & 6 \\ & \downarrow & & & \\ & & 2 & -4 & -10 \\ \hline & 1 & -2 & -5 & -4 \end{array}$$

- 2 Interpret the result.

Quotient: $x^2 - 2x - 5$

Remainder: -4

- 3 **Conclusion.**

Since the remainder is not zero, 2 is not a zero of $f(x)$. The quotient polynomial is $x^2 - 2x - 5$, whose zeros can be found by solving $x^2 - 2x - 5 = 0$ using the quadratic formula.

Example 2

Given that -1 is a zero of $f(x) = 2x^3 + x^2 - 4x - 4$, use synthetic division to find the remaining zeros.

- 1 Set up synthetic division with $r = -1$.

$$\begin{array}{r|rrrr} -1 & 2 & 1 & -4 & -4 \\ & \downarrow & & & \\ & & -2 & 1 & 3 \\ \hline & 2 & -1 & -3 & -1 \end{array}$$

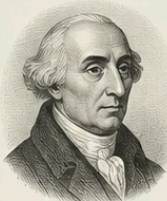
- 2 Interpret the result.

Quotient: $2x^2 - x - 3$

Remainder: -1

- 3 **Conclusion.**

Since the remainder is not zero, -1 is not a zero. Solve the quadratic equation $2x^2 - x - 3 = 0$ to find the remaining zeros. (It factors as $(2x - 3)(x + 1) = 0$, so the zeros are $x = \frac{3}{2}$ and $x = -1$.)



A Bit of History

Joseph-Louis Lagrange (1736–1813), one of the great mathematicians of the 18th century, popularized synthetic division as an efficient method for dividing polynomials.

Key Takeaways

- If r is a zero, then $x - r$ is a factor.
- Synthetic division is a fast and organized way to divide by $x - r$.
- The quotient has one degree less than the original polynomial.
- Use the quotient to find the remaining zeros.

Poster correction: Both “given zero” assumptions on this poster are false for the polynomials printed. The first division gives $P(x) = (x - 2)(x^2 - 2x - 5) - 4$, so $P(2) = -4$. The second gives $P(x) = (x + 1)(2x^2 - x - 3) - 1$, so $P(-1) = -1$. The division remainders are useful, but the quotient roots cannot be called the original polynomial’s “remaining zeros.” Use the verified, different examples below.

Image 5 practice: Reduce the polynomial with synthetic division

Q21. Given that 1 is a zero of $x^3 + 2x^2 - 13x + 10$, divide synthetically and find all zeros.

Q22. Divide $x^3 - 4x^2 - 9x + 36$ by $x + 3$, then find all zeros.

Q23. Use synthetic division with $r = \frac{1}{2}$ to factor $2x^3 - 3x^2 - 5x + 3$ completely.

Q24. Given that 2 is a zero of $x^4 - 2x^3 + 9x^2 - 18x$, divide by $x - 2$ and find all real and nonreal zeros.

Q25. Use synthetic division twice to show that 3 is a repeated zero of $x^4 - 5x^3 + x^2 + 21x - 18$. Then find all zeros and their multiplicities.

Zeros and Their Multiplicities

Understand how repeated roots (multiplicities) affect the graph of a polynomial at the x -axis.

Key Idea

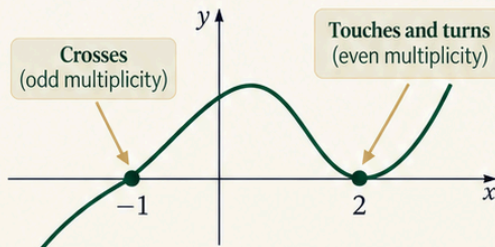
If a polynomial has a zero at $x = a$ with multiplicity m , then $(x - a)^m$ is a factor of the polynomial.

- If m is odd, the graph crosses the x -axis at $x = a$.
- If m is even, the graph touches the x -axis and turns around at $x = a$.

Example 1

Consider $f(x) = (x - 2)^2(x + 1)$.

- Zeros: $x = 2$ (multiplicity 2),
 $x = -1$ (multiplicity 1)
- At $x = 2$, the graph touches the x -axis and turns around (even multiplicity).
- At $x = -1$, the graph crosses the x -axis (odd multiplicity).

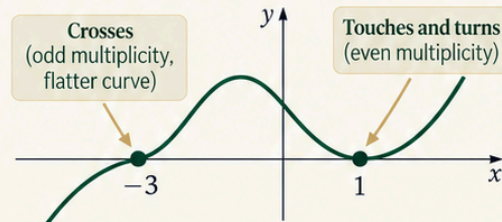


Zeros: -1 (multiplicity 1), 2 (multiplicity 2)

Example 2

Consider $g(x) = (x + 3)^3(x - 1)^2$.

- Zeros: $x = -3$ (multiplicity 3),
 $x = 1$ (multiplicity 2)
- At $x = -3$, the graph crosses with a flattened curve (odd multiplicity greater than 1).
- At $x = 1$, the graph touches and turns (even multiplicity).



Zeros: -3 (multiplicity 3), 1 (multiplicity 2)



Why It Matters

The multiplicity of a zero tells us how the graph behaves at the x -axis, which helps us sketch polynomial graphs and understand their shape.

Key Takeaways

- ✓ A zero with even multiplicity: graph touches and turns at the x -axis.
- ✓ A zero with odd multiplicity: graph crosses the x -axis.
- ✓ Higher odd multiplicities cross with a flatter curve.
- ✓ The multiplicity comes from the exponent of the corresponding factor $(x - a)^m$.

Sketch-reading note: For the first displayed example, $(x - 2)^2(x + 1)$, the degree is three with a positive leading coefficient, so the far-left tail must point down. Treat the poster sketches as illustrations of local crossing/touching, not precise full plots. "Higher multiplicity crosses" applies only when that multiplicity is odd.

Image 6 practice: Read multiplicity at each zero

Q26. For $f(x) = (x + 4)^2(x - 3)^3$, list each zero, its multiplicity, and whether the graph crosses or touches.

Q27. Analyze the zeros of $g(x) = -2(x - 5)^4(x + 2)$. Does the factor -2 change their multiplicities?

Q28. For $P(x) = -(x + 6)^3(x - 1)^2$, identify the sign immediately to the left and right of each zero.

Q29. A polynomial crosses with multiplicity three at -2 , touches with multiplicity two at 1 , and passes through $(0, 16)$. Find the least-degree polynomial and its degree.

Q30. A real polynomial has exactly three distinct real zeros: it touches at -5 , crosses with flattening at 0 , and touches at 4 . What is its least possible degree? Construct one with leading coefficient -1 and justify why degree six is impossible.

End Behavior of Polynomial Functions

Use the degree and leading coefficient to determine the behavior of a polynomial function as $x \rightarrow \infty$ and $x \rightarrow -\infty$.

Key Idea

For a polynomial $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$, the end behavior is determined by the leading term $a_n x^n$.

- If n is even, both ends of the graph go in the same direction.
- If n is odd, the ends go in opposite directions.

End Behavior Summary

Degree (n)	Leading Coefficient (a_n)	As $x \rightarrow -\infty$	As $x \rightarrow \infty$
Even	$a_n > 0$	$f(x) \rightarrow \infty$	$f(x) \rightarrow \infty$
Even	$a_n < 0$	$f(x) \rightarrow -\infty$	$f(x) \rightarrow -\infty$
Odd	$a_n > 0$	$f(x) \rightarrow -\infty$	$f(x) \rightarrow \infty$
Odd	$a_n < 0$	$f(x) \rightarrow \infty$	$f(x) \rightarrow -\infty$

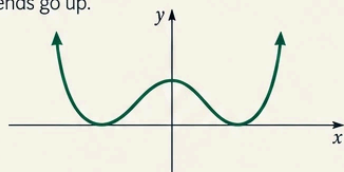
Example 1

Determine the end behavior of $f(x) = 2x^4 - 3x^2 + 1$.
The degree is 4 (even) and the leading coefficient is $2 > 0$.

As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

Both ends go up.



Even degree, positive leading coefficient
(both ends up)

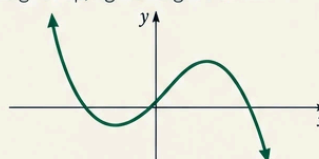
Example 2

Determine the end behavior of $g(x) = -x^3 + 4x$.
The degree is 3 (odd) and the leading coefficient is $-1 < 0$.

As $x \rightarrow -\infty$, $g(x) \rightarrow \infty$

As $x \rightarrow \infty$, $g(x) \rightarrow -\infty$

Left end goes up, right end goes down.



Odd degree, negative leading coefficient
(left up, right down)



Key Takeaways

- End behavior depends only on the degree and leading coefficient.
- Even degree: both ends go in the same direction.
- Odd degree: ends go in opposite directions.
- The lower-degree terms do not affect the end behavior.

Real-World Connection



Polynomial functions are often used to model real-world situations, such as the shape of a roller coaster track. The end behavior helps us understand what happens to the model for very large positive or negative values of x .

Poster sketch note: The four-case table is the rule to use. The small sketches illustrate tail directions but should not be used to read the actual zeros of the displayed example equations.

Image 7 practice: Determine end behavior

Q31. Describe both tails of $f(x) = -5x^6 + 2x^3 - x + 9$.

Q32. Describe both tails of $g(x) = 3x^5 - 7x^2 + 4$.

Q33. Without expanding, determine the degree, leading coefficient, and end behavior of $-3(2x - 1)^2(1 - x)^3$.

Q34. For $P_a(x) = ax^4 - 2x^3 + 5$, describe both tails when $a > 0$, when $a < 0$, and when $a = 0$.

Q35. Determine the actual degree, leading coefficient, and end behavior of $P(x) = (x^2 + 1)(x^2 - 4) - x^4 + 2x^3$. Explain why calling it a quartic would give the wrong tail pattern.

Connecting Zeros, Factors, and Graphs

Use the zeros (and their multiplicities), the degree, and the leading coefficient to sketch the graph of a polynomial.

Key Idea

The zeros (and their multiplicities) tell us where the graph crosses or touches the x -axis. The degree and leading coefficient tell us the end behavior. Combining this information helps us sketch the graph.

Example

Consider $f(x) = -(x + 1)(x - 2)^2(x - 4)$.

1 Zeros and multiplicities

$x = -1$ (multiplicity 1) $\rightarrow$ crosses
 $x = 2$ (multiplicity 2) $\rightarrow$ touches and turns
 $x = 4$ (multiplicity 1) $\rightarrow$ crosses

2 Degree and leading coefficient

The degree is $1 + 2 + 1 = 4$ (even).
 The leading coefficient is negative (because of the minus sign in front).

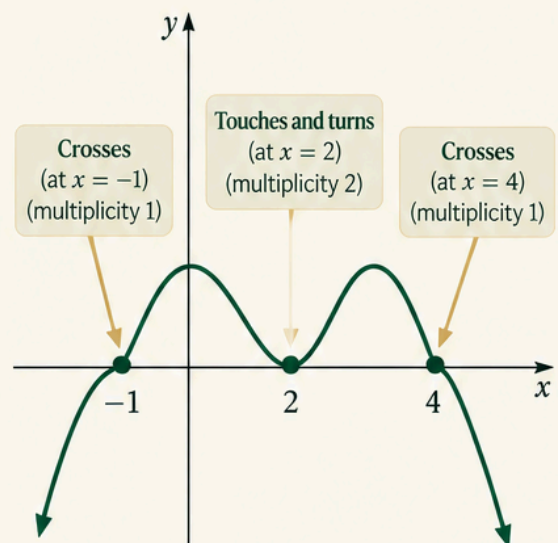
3 End behavior

As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

Both ends go down.

Graph



Key Takeaways

- ✓ Each zero tells us where the graph crosses or touches the x -axis.
- ✓ Even multiplicity: graph touches and turns.
- ✓ Odd multiplicity: graph crosses.
- ✓ The degree and leading coefficient determine the end behavior.
- ✓ Combining all this information helps us sketch the graph of the polynomial.

Real-World Connection



Polynomial functions are used to model real-world situations, such as the shape of a roller coaster track. The zeros tell us where the track hits the ground, and the multiplicities and end behavior help us understand how the track moves between these points.

Image 8 practice: Connect zeros, factors, and graphs

Q36. For $g(x) = (x - 4)^2(x + 2)$, find the zeros, multiplicities, y -intercept, and both tail directions.

Q37. Describe the key graph features of $h(x) = -(x + 5)(x - 2)^2(x - 1)$.

Q38. For $P(x) = (x + 4)(x - 1)^2(x - 5)$, find the sign on each interval separated by its zeros. State which zeros cause sign changes.

Q39. Find all zeros of $P(x) = 2x^3 - 7x^2 - 7x + 12$, then give its intercepts and end behavior.

Q40. A least-degree polynomial crosses at -4 , touches at 1 , crosses at 3 , and has $P(0) = 24$. Find it, state its end behavior, and identify all intervals on which it is positive.

Homework review: R1–R5

R1. For $P(x) = x^2 - 9x + 20$, check whether $x = 4$ is a zero. State the corresponding point on the graph and a factor of $P(x)$.

R2. Find every zero of $P(x) = x^3 - 7x^2 + 10x$ by factoring completely.

R3. List all possible rational zeros of $P(x) = 3x^3 - 5x^2 + 2x - 8$. Does the list prove that any candidate is a zero?

R4. For $P(x) = x^3 - 4x^2 - x + 4$, test $x = 1$ and $x = 2$. Which corresponding linear factor divides $P(x)$ exactly?

R5. Given that -1 is a zero of $P(x) = x^3 - 2x^2 - 13x - 10$, use synthetic division by $x + 1$ and find every zero.

Homework review: R6–R8

R6. For $f(x) = (x + 4)^3(x - 1)^2$, state each zero and its multiplicity. At which zero does the graph cross the axis, and at which does it touch and turn?

R7. Describe the end behavior of $f(x) = -3x^6 + 2x^4 - 7x + 9$. Identify the term that determines your answer.

R8. For $f(x) = -(x + 2)(x - 1)^2(x - 5)$, find the zeros, crossing or touching behavior, y-intercept, and both tail directions.

Homework word problems: W1–W5

W1. A game's score difference from its target is $D(t) = t - 7$ points after t minutes, for $0 \leq t \leq 10$. A negative difference means below the target. When does the score reach the target, and what does this say about the graph of D ?

W2. A moving light's signed displacement from a reference line is $d(t) = (t - 1)(t - 4)$ centimeters, for $0 \leq t \leq 5$ seconds. When is the light on the line? Is it above or below the line at $t = 2$, if positive displacement means above?

W3. A school-fair craft booth models its profit by $P(n) = -n^2 + 26n - 120$ dollars when it sells n bundles, where n is a whole number from 0 to 26. Find both break-even sales levels and the whole-number sales levels that produce a profit.

W4. A ball tossed upward from a platform has modeled height $h(t) = -5t^2 + 20t + 25$ meters after t seconds. Use the model only from launch until the ball reaches the ground. Find when it lands and when it returns to its launch height.

W5. A rectangular garden has width $x + 1$ meters and length $x + 5$ meters. Its area must be 45 square meters. Write a polynomial whose zero determines x , solve it, and give the actual dimensions.

Homework word problems: W6–W10

W6. A shipping box has inside dimensions x , $x + 1$, and $x + 2$ centimeters and volume 120 cubic centimeters. Show that $x = 4$ is a rational zero of the volume-difference polynomial. Divide out that factor and explain why it gives the only real value of x .

W7. A sensor's calibration error is modeled by $E(s) = 2s^3 - 9s^2 + 7s + 6$, where its control setting must satisfy $0 \leq s \leq 4$. A zero error means the sensor is calibrated. Use the Rational Root Theorem to justify testing $s = 2$, then find every calibrated setting the device can actually use.

W8. During a short inspection, a drone's signed height relative to a horizontal reference line is $d(t) = (t - 2)^2(t - 5)$ meters, for $0 \leq t \leq 6$. Negative values mean below the line, not below the ground. When does it meet the line? At which meeting does it pass through the line, and where is it relative to the line at $t = 3$?

W9. A designer models the height of a ramp above the floor by $h(x) = a(x + 1)(x - 3)^2$, using only $0 \leq x \leq 4$ meters. The height at $x = 0$ is 9 meters. Find a , determine where the actual ramp meets the floor, and find its modeled height at $x = 4$.

W10. For a limited production range $1 \leq x \leq 7$, a factory models profit by $P(x) = -x^3 + 12x^2 - 44x + 48$, in thousands of dollars, where x is production in hundreds of units. Verify that $x = 2$ is a zero, find all break-even output levels, and identify the intervals with positive profit within the stated range.

Homework word problems: W11–W12

W11. A temporary shop models weekly profit by $P(w) = -10(w + 1)(w - 2)^2(w - 6)$ dollars for $0 \leq w \leq 8$ weeks. Find the break-even times in this interval, decide whether profit changes sign at each, and describe both mathematical tails of the polynomial. Explain why those tails are not a forecast for the shop forever.

W12. A sculpture designer uses a polynomial $F(x)$ for vertical displacement from a reference line. Negative horizontal positions are allowed. The profile crosses the line at $x = -3$ and $x = 5$, touches it at $x = 1$, and passes through $(0, 30)$. Find the least-degree polynomial, describe its end behavior, and decide whether its point at $x = 4$ is above or below the line.